

Is Cybersecurity Safe From AI?

The planning guide — the full scores by track, the six factors behind them, the honest trade-offs, and what to ask before you commit.

5.0

AI EXPOSURE · INCIDENT RESPONSE / THREAT HUNTING

where 10 is most at risk

For anyone still choosing — read it straight through.



Before you read this

You're weighing a cybersecurity degree — or a path into the field — and you want to know what that choice actually looks like in an AI economy. That's the whole job of this guide.

It's written to you directly, whether you are the one choosing or a parent thinking it through alongside them. Where it says "you," a parent can read "them"; it's the same decision from either side.

What this is. A facts document, not a recommendation. It sets out what's known, where the evidence is contested, and where we might be wrong — and then how to choose well given all of it.

If you're the parent, a note on how to use it. Don't lead with the scores. Open a conversation with a risk number and one of two things happens: they shut down, or they get defensive about a choice they're already leaning toward. Read it yourself first, then share it as information you both react to, not a verdict you've reached. They're choosing, not you — the job is to widen what's been considered, not to narrow it.

And one thing specific to cybersecurity. This is the field with the largest gap between how it is marketed and how it actually hires. You've almost certainly encountered the figure of 4.8 million unfilled cybersecurity jobs — it appears in university prospectuses, bootcamp advertising and certification marketing everywhere. The number is real. What it means for a beginner is not what those sources imply, and the section on that figure explains why. That gap is the single most useful thing in this document.

The short answer

Better than most technical paths, and not as protected as the marketing suggests.

Incident response and threat hunting scores **5.0** out of 10. Tier-one SOC analyst work scores **7.4**.

Cybersecurity is the best-scoring pure-technology track we measure — but "safer than software" and "safe" are different claims. Its floor is 5.0, against 4.0 for licensed engineering and 2.5 for a service electrician.

And the entry route everyone is pointed toward is the specific part being automated — which is why *which version you aim at* is most of the decision in front of you.

The scores

ROLE	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Incident response / threat hunting	4.2	4.6	5.0	+0.8	Moderate
Security architecture	4.7	5.1	5.5	+0.8	Moderate
Penetration testing / offensive	4.9	5.3	5.7	+0.8	Moderate
GRC / compliance	5.2	5.7	6.1	+0.9	Moderate-High
SOC analyst (tier 1)	6.1	6.8	7.4	+1.3	High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, entry-level software development scores 8.1, entry data analysis 8.2, and licensed engineering 4.0.

Read the table this way: nothing in cybersecurity scores below 5.0. It genuinely is the best position in technology, and it is a mid-table position in this index overall.

Why cybersecurity is protected — the six factors



Here's how incident response and threat hunting answers them. Knowing *why* the entry SOC rung scores 7.4 while incident response holds 5.0 is what tells you which version of the field to aim at.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

Most of tier-one triage, and very little above it.

Alert triage and false-positive filtering. Log correlation and enrichment. Vulnerability scanning and reporting. Standard playbook execution. Compliance evidence collection. Phishing detection and classification.

And the projection here is unusually specific. Gartner projects that more than 50% of SOC Tier 1 analyst responsibilities will be handled by AI by 2028. That is a named forecast about the exact role most people are told to start in.

What resists: deciding whether this is an incident or noise, working out what an adversary is actually trying to do, judgment when the attack has no precedent, making the call to shut down production, and explaining a breach to a board or a regulator.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Harder than the marketing suggests, and the section on the 4.8 million figure covers why in detail.

The short version: the shortage is for experienced people. Practitioners consistently report that "entry level" listings ask for two years of experience, and ISC2's 2025 workforce study found budget constraints have overtaken talent shortage as the number one hiring barrier.

The way through isn't to wait for the entry rung to soften — it may not. It's to enter beside it: build demonstrable, hands-on evidence before applying — home labs, capture-the-flag competitions, bug bounty work — because in a field with no license that is the credential; aim placements at incident response, security engineering or offensive work rather than pure SOC monitoring; and use apprenticeship or degree-apprenticeship routes, which are paid and solve the experience paradox directly. Those are concrete moves, not a hope — and choosing a program that makes them possible is what the second half of this guide is about.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Almost none of it. Security work is fully digital, which removes one protection entirely.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

At the incident and architecture level, substantially.

Someone has to make the call to take a system offline, and someone has to answer to a board or a regulator afterwards. That accountability attaches to a person.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Weakly. There is no cybersecurity license, though regulatory regimes increasingly require named accountable individuals for security posture — which is a partial and growing protection.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

We rate this 9.0, and the section on adversarial novelty explains why cybersecurity's novelty is a different kind from every other career in this index.

The 4.8 million number, and what it actually means

This is the most useful section in the guide, because the claim it examines is everywhere.

The numbers are real:

- ISC2's Workforce Study puts the global gap at roughly **4.8 million unfilled positions**
- CyberSeek — a joint effort by CompTIA, Lightcast and NIST — puts the US figure at around **700,000–750,000**
- The World Economic Forum estimates the workforce needs to grow **87%** to meet demand
- BLS projects information security analyst employment growing **29–33% through 2034**, roughly seven times the average across all occupations

Every one of those is accurate. And a beginner should read them carefully.

First: the shortage is for experience, not for people. The consistent report from practitioners and hiring data alike is that "entry level" postings ask for two years of experience. Organisations want people who can direct and challenge security work, not people who need training. The gap persists precisely because new entrants aren't being developed.

Second: budget has overtaken talent as the constraint. ISC2's 2025 study found budget cuts displacing talent shortage as the primary hiring barrier. A vacancy that exists on paper but has no funded headcount behind it does not employ anyone.

Third, and most directly relevant to our score: the entry route being marketed is the one being automated. Gartner's projection that more than half of SOC Tier 1 responsibilities will be AI-

handled by 2028 lands squarely on the job that bootcamps, certifications and university programs point graduates toward.

Fourth, the pipeline problem this creates. One industry analysis puts it well: a workforce with no entry-level pipeline produces no tier-two analysts and no senior generalists in five years. The same argument we found in law — the sector may be making a self-defeating mistake — applies here, and nobody has resolved it.

Where this leaves a student. Cybersecurity is a genuinely good career with genuine demand. It is not the frictionless six-figure on-ramp it is marketed as, and the specific route being sold is the weakest one. The AI-native section covers what to do instead.

Adversarial novelty — a protection nothing else in this index has

We rate threat hunting 9.0 on novelty, and it is worth understanding why, because the mechanism is unique.

AI is strong on patterns and weak on genuine novelty. Most careers earn a high rating here through complexity — a one-off engineering site, an atypical patient, a case with no precedent.

Cybersecurity earns it because someone is deliberately manufacturing novelty against you.

That distinction matters more than it sounds. An unusual medical presentation is unusual by accident. An unusual attack is unusual *on purpose* — designed specifically to look benign, or to exploit the gap between what a detection system was trained on and what is actually happening.

This creates a property no other profession in this index has: **the novelty is adaptive**. Whatever pattern a defensive system learns to recognize, an adversary has an incentive to stop matching it. There is no equilibrium where the patterns settle down and the work becomes routine, because the other side is paid to prevent exactly that.

That is why threat hunting holds 5.0 despite no physical component, no license, and a fully digital workflow. Almost nothing else in this index survives that combination of missing protections.

But notice where the protection does not reach. A tier-one SOC analyst is not facing adversarial novelty. They are working a queue of alerts against a playbook — pattern-matching, in a role explicitly designed to be pattern-matching. That is why it rates 4.5 rather than 9.0, and why it scores 7.4.

The general lesson: adversarial work resists automation better than merely complex work. Complexity can be learned. An opponent who changes when you learn cannot be. Any career with a genuine adversary — security, litigation, fraud investigation, negotiation, competitive strategy — carries a protection that complexity alone does not provide.

What the trend shows

SOC analyst tier 1: 6.1 → 6.8 → 7.4. Up 1.3.

Incident response: 4.2 → 4.6 → 5.0. Up 0.8.

The gap widened from 1.9 points to 2.4. The distinctive feature here is that both ends moved, and the protected end moved faster than protected ends usually do — because cybersecurity has no license or physical requirement holding it still, only judgment and adversarial dynamics.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Judgment under novelty	Strongest available	Most durable here	Adaptive adversaries prevent patterns from settling. Uniquely resistant.
Human trust / accountability	Real at incident level	Durable	Someone must decide to take production down and answer for it.
Regulatory / licensing	Weak but growing	Strengthening slowly	Regulation increasingly requires named accountable individuals.
Physical / embodied	None	n/a	Fully digital work.

The honest read. Cybersecurity's protection is unusual: it comes almost entirely from the adversary rather than from structure. That is genuinely durable — as long as the adversary exists, which is not in doubt — but it protects the roles that face the adversary directly, and not the ones processing alerts behind them.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Triaging the alert queue	Reviewing what the system escalated
Correlating logs by hand	Interrogating automated correlation
Running the vulnerability scan	Judging which findings actually matter here
Following the playbook	Deciding when the playbook doesn't apply
Learning the environment by working the queue	Learning it... some other way, unresolved
Deciding to shut down production	Unchanged

The fifth row is the recurring problem across this index, and cybersecurity has it acutely. Analysts learned an organisation's environment — what normal looks like — by working thousands of alerts. That was tedious, and it was how you developed the instinct to know when something is wrong. Nobody has replaced it.

And there's a second-order effect worth naming. AI improves both sides simultaneously. This is the only career we score where the same technology strengthens the attacker and the defender at once, which is part of why demand holds even as tier-one work automates.

Human strengths that will matter most

1. **Adversarial thinking** — modeling what someone actively hostile would do. The core skill, and the hardest to teach. 2. **Deciding under pressure with incomplete information** — most incidents are ambiguous at the moment the decision has to be made. 3. **Accountability** — being the person who takes production down, and answers for it. 4. **Explaining to non-technical people** — boards, regulators, executives. Increasingly the difference between a good analyst and a security leader. 5. **Verification** — knowing when an automated detection is wrong in either direction.

Notably absent: tool fluency, alert throughput and certification breadth. These were the traditional markers of a strong junior and are the fastest-depreciating.

And underneath all five, two traits the score can't grade: curiosity and adaptability. The specific tools — the SIEM, the EDR platform, the framework of the moment — will turn over many times across a career, and so will the attacks themselves. The people who compound are the

ones who stay genuinely curious about how systems break and re-tool fastest, rather than defending the one stack they learned. In a field with no license and a permanently moving adversary, that flexibility is itself a form of protection — and increasingly it's what employers are hiring for: the trajectory toward judgment and investigation, not the current certification count.

Other things to consider about this job

What's genuinely good about it

Demand is real and structurally driven. BLS projects 29–33% growth through 2034 — roughly seven times the average. Ransomware, breach frequency and regulatory obligations are all increasing, and none of those is cyclical.

Entry pay is solid: SOC and GRC analyst roles report roughly \$70,000–\$85,000 depending on location, junior incident response \$70,000–\$90,000, and junior penetration testing \$75,000–\$95,000.

Entry routes are unusually flexible. ISC2's hiring research found around 90% of organisations will consider candidates with only IT experience, and 89% accept entry-level certifications in place of a degree. Very few professional fields are that open.

What practitioners report as rewarding: the work genuinely matters when it matters; there is a real opponent, which some people find energizing in a way ordinary problem-solving isn't; continuous learning is intrinsic rather than optional; and progression to well-paid senior and leadership roles is fast for people who stay.

What's genuinely hard about it

Burnout is severe and unusually well documented.

- Bitsight's 2025 survey of over 1,000 risk and security professionals found 47% reporting some level of burnout
- Proofpoint's Voice of the CISO 2025 found 63% of CISOs had experienced or witnessed burnout in the past year
- Sophos measured 4.8 hours per week lost to burnout, up more than 25% year on year
- Splunk's State of Security, surveying 2,058 practitioners, found 52% of security teams overworked

- Industry analysis finds attrition exceeds other technology roles

On-call is a permanent feature, not an early-career phase. Incidents happen at night and at weekends.

The certification treadmill is real and never stops, and much of it is self-funded.

And the entry paradox is the practical obstacle: the field wants experience, and the traditional way of getting it is being automated.

Who this work suits — and who it doesn't

Cybersecurity tends to suit people who:

- **Think adversarially** — instinctively asking how something could be abused rather than how it should work. The defining trait, and largely untrainable.
- **Stay calm when things are going badly** — incidents are the job
- **Are comfortable with ambiguity** and deciding without complete information
- **Enjoy continuous learning** rather than tolerating it
- **Can explain technical things to people who don't want the detail** — this is what separates analysts from leaders

It's harder for people who:

- Need **predictable hours** — on-call is structural
- Want **defined, completable tasks** — security is never finished
- Find **sustained pressure** corrosive rather than energizing
- Chose it because it was marketed as the **safe tech career**. That framing is what the 4.8 million section is about.

What else is moving this market

Regulation is a genuine tailwind. Compliance obligations create demand that has nothing to do with attack volume, and increasingly require named accountable individuals — which is a protection our regulatory factor is only beginning to reflect.

Retention is a bigger drain than layoffs. ISACA's research found organisations struggling to retain security staff even without cutting — voluntary turnover driven by burnout is a larger problem than involuntary loss, which is why senior people remain scarce.

And AI is being adopted cautiously. Splunk found 33% of teams planning to fill skills gaps with AI and automation, but only 11% trusting AI completely for mission-critical tasks. That gap between adoption and trust is where the human roles sit.

The AI-native advantage — what to actually do about it

The situation: security teams are adopting AI faster than they trust it. Only about one in nine trusts it completely for mission-critical work, while a third plan to use it to fill skills gaps. That gap is the opportunity. A good program should be building this fluency deliberately, not banning the tools or ignoring them.

WHAT AN AI-NATIVE SECURITY PROFESSIONAL LOOKS LIKE

1. Verify. Knowing when an automated detection is wrong — in either direction. A false negative is a breach; a false positive at scale is alert fatigue, which is itself a security failure.

2. Think adversarially about AI itself. Attackers are using the same tools. Understanding how to attack and defend AI systems is a specialism that barely existed three years ago and is growing fast.

3. Move up the escalation chain deliberately. Tier one is automating. Every step toward investigation, hunting and architecture moves someone from a 7.4 to a 5.0.

4. Own a decision. Everything in this guide points at accountability — being the person who calls it.

5. Communicate. Explaining an incident to a board is the skill that turns an analyst into a leader, and it is chronically undersupplied.

CONCRETE PREPARATION

Before the degree: build hands-on evidence — home labs, capture-the-flag competitions, bug bounty work, contributing to open-source security tooling. In a field with no license, demonstrable capability is the credential, and it is available to a sixteen-year-old for free.

During: get Security+ or an equivalent early, then specialize. Aim placements at incident response, security engineering or offensive work rather than pure SOC monitoring.

Entering: take the SOC role if it's the way in, but treat it as a two-year staging post rather than a career, and be deliberate about moving to tier two and beyond.

The routes in — and the versions that hold up

ROUTE	ASSESSMENT
Cybersecurity degree with real lab depth	Works if technical rather than compliance-focused.
Computer science + security specialization	Often stronger. The engineering foundation is what verification depends on.
IT or networking role → security	Genuinely common and respected. Around 90% of organisations consider candidates with only IT experience.
Apprenticeship / degree apprenticeship	Expanding, paid, and solves the experience paradox directly. Under-used.
Certification-only route	Possible — 89% of organisations accept entry certifications in place of a degree — but competitive without hands-on evidence.
Bootcamp aimed at SOC tier one	The weakest position: it optimises for the specific role Gartner projects being half-automated by 2028.

Where a cybersecurity career can lead later

Broad and well-paid: incident response and digital forensics, security architecture, offensive security and red teaming, cloud and application security, AI/ML security, governance risk and compliance, security leadership and CISO roles, and consultancy.

The pattern in this index holds, with a security-specific twist: roles facing the adversary directly score better than roles processing what the adversary triggered. Cloud security, AI/ML defense and application security are named consistently as the most in-demand skills, and all three sit on the protected side of that line.

What to look for in a cybersecurity program

1. Lab time and hands-on depth. How many hours in actual environments, breaking and defending real systems? This is the single best differentiator.

2. Does it teach adversarial thinking, or tool operation? A curriculum organised around products is training for the automating tier.

3. Specialization pathways. Incident response, offensive security, cloud, application security. Generic security degrees place worst.

4. Employer relationships and apprenticeship routes. The experience paradox is real; programs that solve it are worth more than ranked ones.

5. First-destination job titles — SOC analyst versus security engineer is a meaningful difference.

Questions to ask a cybersecurity program

On practice:

- How many hours do students spend in hands-on lab environments?
- Do students participate in competitions or run real security exercises?
- Is there an apprenticeship or placement route?

On the curriculum:

- Is the curriculum organised around adversarial thinking or around specific tools?
- Which specializations are available, and how many students take each?
- How does the program address AI — both as a defensive tool and as an attack surface?

On outcomes:

- What job titles do graduates hold at six months?
- What proportion are in SOC roles versus engineering, IR or offensive positions?

Red flags: a curriculum built around vendor tools; minimal lab time; no specialization pathways; outcomes quoted as "employed in cybersecurity"; heavy emphasis on the 4.8 million figure in the marketing.

The career-hunting checklist

Choosing well is step one. This is what turns a good choice into a good outcome once you're in — worth reading now, because the best time to pick a program is knowing exactly what you'll need to do with it. The first list is for the degree and training years; the second is for the job hunt that follows.

During the degree / training — build what can't be automated

- **Aim at a protected specialization, not the SOC queue.** Incident response and threat hunting (5.0), security architecture (5.5) and offensive security / penetration testing (5.7) are the protected end; tier-one SOC monitoring (7.4) is the exposed one. Point every elective and placement at the former.
- **Build hands-on evidence before anyone asks for it.** Home labs, capture-the-flag competitions, bug bounty work, contributing to open-source security tooling. In a field with no license, demonstrable capability is the credential — and it's free to a sixteen-year-old tonight.
- **Get Security+ or an equivalent early, then specialize.** The baseline cert opens the door; depth in one protected track is what carries you through it. Certification breadth alone is among the fastest-depreciating things you can collect.
- **Chase the apprenticeship or placement route.** Degree apprenticeships are paid and solve the experience paradox directly — the single hardest problem in this field — and they're badly under-used.
- **Get fluent with AI as a defender, not a shortcut.** Learn to verify an automated detection in both directions, and learn how to attack and defend AI systems themselves — a specialism that barely existed three years ago.
- **Feed curiosity and adaptability on purpose.** The tools and the attacks both turn over constantly; the person who re-tools fastest, and stays genuinely interested in how systems break, compounds.
- **Keep the frame wide.** Vet computer science with a security specialization, apprenticeship routes, and adjacent technical paths properly before committing. The most crowded lane is the SOC-tier-one bootcamp, and it points at the automating role.

During the job hunt — aim, don't just apply

- **Target the protected lanes on purpose.** Incident response, security engineering, offensive work, cloud and application security — not "SOC analyst, open to anything," which points straight at the 7.4.

- **Read the title, not just the offer.** "Tier-one SOC analyst" is the automating layer. If you take one, take it as a deliberate two-year staging post with a move to tier two and beyond in mind, not as the plan.
- **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. An employer still running a large tier-one SOC intake may simply be behind the curve — and that's exactly the rung Gartner projects half-automated by 2028. Availability can be a lagging indicator, not a green light.
- **Prefer the seat that faces the adversary.** A role where you investigate, hunt or architect — where someone is deliberately manufacturing novelty against you — carries the protection nothing else in this index has. Proximity to the adversary is proximity to the protection.
- **Ask the employer where AI is heading in the role.** Whether they're building the job around investigation and verification, or waiting to be forced to, tells you if the seat grows or is quietly being hollowed out.
- **Weigh trajectory over the starting salary.** Entry pay is solid across the field — SOC and GRC \$70,000–\$85,000, junior incident response \$70,000–\$90,000, junior penetration testing \$75,000–\$95,000 — so choose the lane that ages best, not the highest first-year number in an automating one.

Go deeper — further reading

- **Information Security Analysts — Occupational Outlook Handbook** — US Bureau of Labor Statistics. The primary source for the growth projection and wage data. Free, and the least commercially interested source available on this career.
- **ISC2 Cybersecurity Workforce Study** — the origin of the 4.8 million figure, and worth reading in full rather than via the headline. The 2025 edition's finding that budget constraints have overtaken talent shortage as the top hiring barrier is the part almost nobody quotes.
- **CyberSeek** — CompTIA, Lightcast and NIST. An interactive map of actual US cybersecurity demand by location and role. Useful for seeing which roles are genuinely open near you rather than in aggregate.

The counterargument — read this one:

The strongest case against our 7.4 for tier-one work is the demand data itself: 4.8 million unfilled positions globally, 29–33% projected growth, and organisations so short-staffed

that 90% will consider candidates with only IT experience. On that reading, automating tier-one triage doesn't eliminate the role — it makes an impossibly under-resourced function survivable, and the analysts move up rather than out.

There's a version of this that is explicitly the industry's plan: redesign entry-level roles around AI augmentation rather than eliminating them, precisely because a workforce with no entry pipeline produces no senior generalists in five years.

Where we land: we think the demand is real and the pipeline argument is sound — and that the same argument was made loudly in law for two years while graduate hiring moved the other way. Intent and behavior are different things. We score the market as it is.

The specific thing we're watching: whether tier-one SOC postings hold or fall as the Gartner 2028 horizon approaches. If organisations genuinely redesign entry roles around augmentation rather than reducing them, our 7.4 is too high and we will lower it.

Bottom line

Cybersecurity is the best position in technology and a mid-table position overall — and the gap between those two statements is where most of the confusion lives. That's the good news for someone still choosing: the decision in front of you — which version, which program, which route in — is exactly the one that determines the outcome.

Its protection is genuinely unusual: an adaptive adversary means the patterns never settle, which is a shield nothing else in this index has. That's why threat hunting holds 5.0 despite having no license, no physical requirement and a fully digital workflow.

But the marketing and the hiring reality have come apart. The 4.8 million figure is real and it describes a shortage of experience, not of people. Budget constraints have overtaken talent as the primary barrier. And Gartner projects more than half of tier-one SOC work handled by AI by 2028 — which is precisely the role that bootcamps, certifications and degree programs point graduates toward.

So the useful advice is specific: build hands-on evidence before applying, because in a field with no license that is the credential and it's free to acquire. Aim at incident response, security engineering or offensive work rather than SOC monitoring. Take a SOC role if it's the way in, but treat it as a two-year staging post rather than a destination. And weigh the burnout data honestly — 47% reporting burnout and attrition above other technology roles is not a footnote.

And one thing worth saying plainly. Cybersecurity is one of very few careers where an adversary guarantees the work stays interesting, and where a sixteen-year-old can start building genuine, verifiable capability tonight for nothing. That combination is rare, and it's a better reason to enter than any workforce statistic.

Questions worth sitting with

Whether you're the one choosing or a parent thinking it through together, these are the conversations that lead to a better decision. Read "you" as whoever the choice is about.

About the work itself

1. What made cybersecurity appealing? Push past "it seems secure" to something specific. 2. **Do you think adversarially?** Do you instinctively ask how something could be abused rather than how it works? That's the defining trait, and it's usually visible early. 3. What does a Tuesday look like in the job you're imagining? What about a Tuesday at 2am during an incident? 4. How do you behave when something is going badly — calmer, or more scattered?

Reacting to the findings

5. Incident response scores 5.0 and tier-one SOC 7.4. Does that change where you'd aim? 6. **Had you encountered the 4.8 million figure?** Where — and does the section on it change how you read it? 7. Gartner projects over half of tier-one SOC work AI-handled by 2028. What do you make of that, given it's the standard entry route?

The harder conversation

8. **The burnout data.** 47% reporting burnout, attrition above other tech roles, on-call as a permanent feature. Have you looked at that squarely? 9. Every "entry level" posting wants two years of experience. What's the plan for getting round that? 10. Have you checked whether the programs you're considering point at SOC roles or at engineering and IR?

Testing it against reality

11. **The most valuable single action, available tonight and free:** set up a home lab, or start capture-the-flag competitions. Nothing else demonstrates capability as directly, and almost

nobody your age has any. 12. Find someone two to three years into a security role and ask what they actually do — and how often they're woken up.

Widening the frame

13. Have you considered **computer science with a security specialization** rather than a cybersecurity degree? The engineering foundation is what verification rests on. 14. What about an **apprenticeship route**? It solves the experience paradox directly and is badly under-used. 15. If cybersecurity weren't available, what would you choose? The answer usually reveals what you're actually drawn to.

This is analysis and informed judgment about a fast-moving situation. For cybersecurity specifically, we've flagged that widely-quoted workforce shortage figures describe a shortage of experience rather than of people, and that the entry route most commonly marketed is the one most directly affected by automation.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Cybersecurity

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned} \text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment}) \end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

INCIDENT RESPONSE / THREAT HUNTING — 5.0 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	3.2	4.1	5.1	1.78
Entry-path erosion	15%	4.0	4.5	5.0	0.75
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	3.5	3.5	3.5	0.65
Judgment under novelty	10%	9.0	9.0	9.0	0.1
				Total	5.01 → 5.0

SECURITY ARCHITECTURE — 5.5 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.3	5.2	6.1	2.13
Entry-path erosion	15%	4.0	4.5	5.0	0.75
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	7.0	7.0	7.0	0.45
Regulatory / licensing	10%	3.5	3.5	3.5	0.65
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	5.49 → 5.5

PENETRATION TESTING / OFFENSIVE — 5.7 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.6	5.5	6.5	2.27
Entry-path erosion	15%	4.2	4.8	5.2	0.78
Physical / embodied	15%	1.5	1.5	1.5	1.27
Human trust / accountability	15%	6.5	6.5	6.5	0.53
Regulatory / licensing	10%	3.0	3.0	3.0	0.7
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	5.71 → 5.7

GRC / COMPLIANCE — 6.1 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.6	5.8	6.7	2.34
Entry-path erosion	15%	4.5	5.2	5.8	0.87
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	6.0	6.0	6.0	0.6
Regulatory / licensing	10%	5.0	5.0	5.0	0.5
Judgment under novelty	10%	5.5	5.5	5.5	0.45
				Total	6.11 → 6.1

SOC ANALYST (TIER 1) — 7.4 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.3	6.9	8.3	2.91
Entry-path erosion	15%	5.5	6.5	7.2	1.08
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	4.5	4.5	4.5	0.82
Regulatory / licensing	10%	3.0	3.0	3.0	0.7
Judgment under novelty	10%	4.5	4.5	4.5	0.55
				Total	7.41 → 7.4

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Incident response / threat hunting

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	3.2	4.0	2.48	4.2
2025	4.1	4.5	2.48	4.6
Fall 2026	5.1	5.0	2.48	5.0

Movement decomposition: automatability contributed +0.66 and entry-path erosion +0.15, for a total of +0.8 points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.